ALMA Proposal Preparation Workshop

March 20, 2026 @IFA-UV, Valparaíso

2.4. ALMA Proposal Review Process

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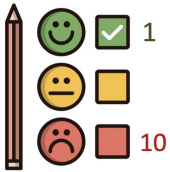
Dual-Anonymous, Distributed Peer Review

Reviewers do not know the PI/team, and proposers do not see the reviewers' names.

Proposals are evaluated by other ALMA proposers who work in a related field.

Both sides remain anonymous

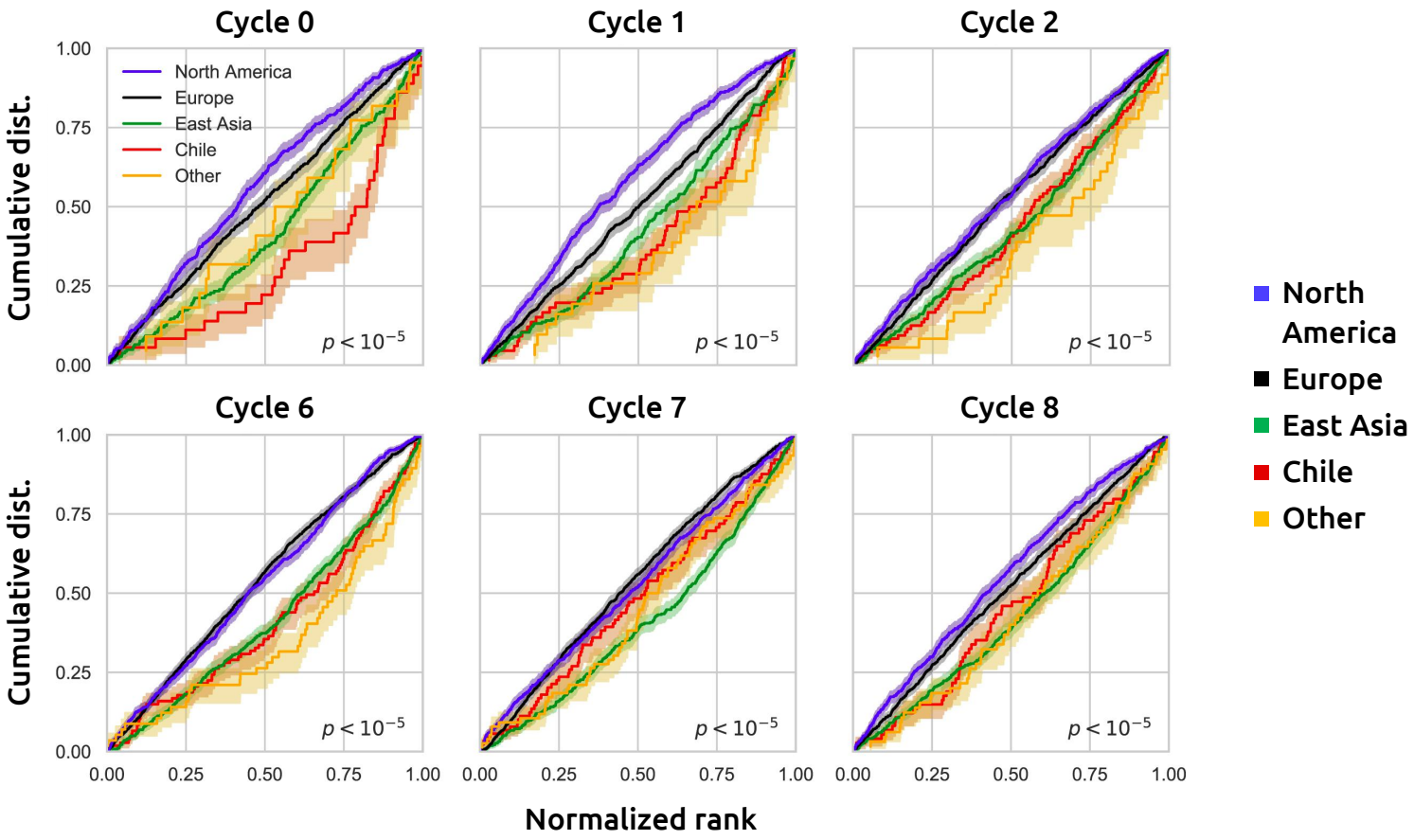
10 reviews per submission*



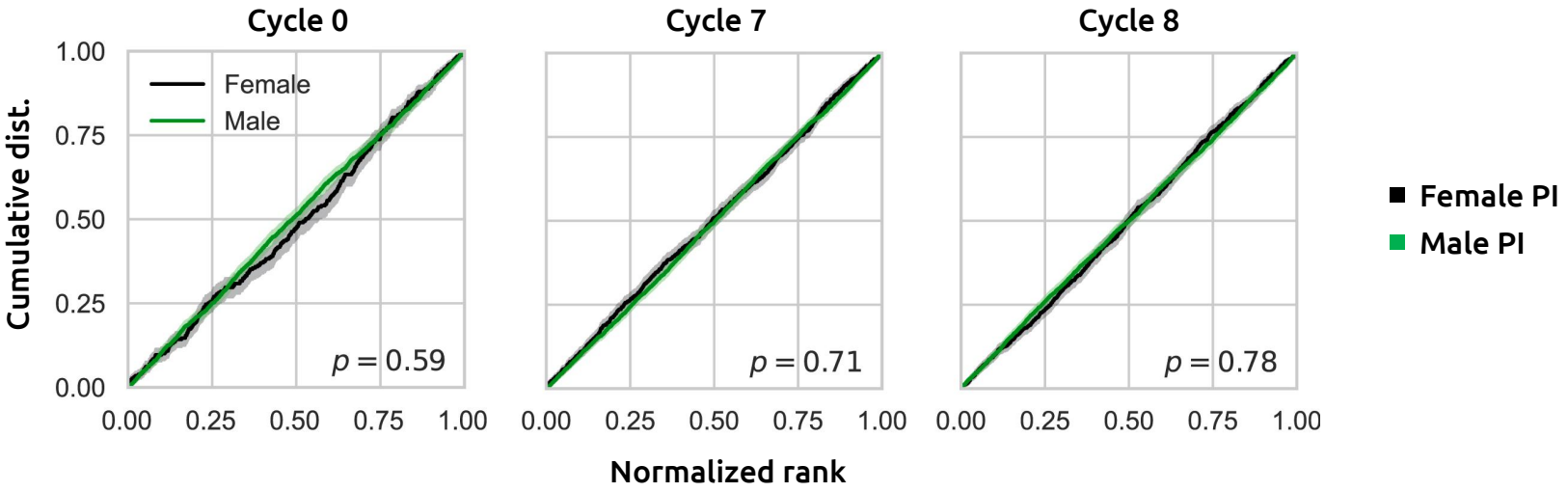
- **Stage I** (mandatory)
  - Declare any conflicts of interest
  - Rank proposals (1–10)
- **Stage II** (optional)
  - Access anonymized reviews
  - Revise comments and rankings

* Excluding Large Programs

Anonymous Review: Ranking Effects



Anonymous Review: Ranking Effects



➤ Following a successful Cycle 7 pilot of dual-anonymous review (Supplemental Call), its adoption in Cycle 8 (2021) has reduced regional and gender disparities.

Reviewer Timeline for Cycle 13

April 23, 2026
Proposal Deadline

1) The PI designates a reviewer in the Observing Tool (OT)
   - ➤ Up to 3 proposal sets (30 proposals) per reviewer

April 28, 2026
Pre-review Stage

1) Reviewer specifies scientific expertise in the ALMA User Account
2) Reviewer declares any conflicts of interest in the Reviewer Tool
3) Deadline to propose an alternative reviewer, if necessary

May 6 – June 3
Review Stage I

1) Plenary sessions on **May 7–11** (attendance highly recommended)
2) Declare any conflicts of interest for assigned proposals by **May 13**
3) Complete all reviews by **June 3** **[Mandatory]**

June 4 – 18
Review Stage II

1) Read reviews from the other reviewers
2) Revise comments and rankings as needed

Slide courtesy of Andrea Corvillón

Conflicts of Interest

- **Assignments are determined based on the reviewer expertise**

  Inferred from proposal history using machine-learning algorithms (Carpenter et al. 2025), and from the expertise keywords specified in the ALMA User Account.

- **Reviewer Tool → Proposal titles & abstracts → Conflicts of interest**

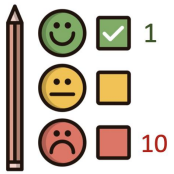
  Definition: The reviewer's work would benefit if the proposal is accepted or rejected.

- **Conflicts are automatically flagged based on:**

  - Current-cycle proposals (reviewer/mentor)
  - Declared conflict-of-interest list
  - Proposal submission history

- **Additional conflicts (not auto-detected) must be declared, e.g.:**

  - Same target and similar science
  - Significant input to a proposal without being listed
  - Any other perceived strong conflict of interest



1 → Strongest
10 → Weakest

Overall scientific merit

- Does the proposal clearly indicate which key scientific questions will be addressed?
- Will the observations address the proposal's science goals and have a high scientific impact in this field? (Will it also benefit closely related fields?)
- Does the proposal clearly describe how the data will be analyzed in order to achieve the science goals? (Are the expected results, uncertainties, and gains quantified?)
- Do **not** consider the requested observing time in your evaluation!

TJ review
No need to be expert: identify inconsistencies

Suitability of the observations to achieve the scientific goals

- Is the choice of target(s) clearly described and well justified?
- Are the requested signal-to-noise ratio, angular resolution, largest angular scale, and spectral setup justified to achieve the science goals?
- Are the observing parameters and quantities consistent throughout the proposal?
- Dupplicate: Does the proposal justify why new observations are needed?
- Joint Proposals: Does the proposal clearly justify the need for multiple facilities?

Learn from other reviewers in Stage II

- Compare the strengths and weaknesses identified by other reviewers
- Revise your comments and rankings if needed
- Although optional, Stage II is crucial: you are rarely an expert in all proposals

[Tip] ESO proposal grading scale

1.0 Outstanding: breakthrough science
1.5 Excellent: definitely above average
2.0 Very good: no significant weaknesses
2.5 Good: minor deficiencies do not detract from strong scientific case
3.0 Fair: good scientific case, but with definite weaknesses
3.5 Rather weak: limited science return prospects
4.0 Weak: little scientific value and/or questionable scientific strategy
4.5 Very weak: deficiencies outweigh strengths
5.0 Unsuitable

Recommended steps for an effective review

➤ Preparation

- **Allocate sufficient time** (Expect 2–3 days per proposal set)
- Mitigate unconscious bias, i.e., judging proposals for non-scientific reasons (often unconsciously, e.g., cultural or language bias, figure aesthetics)
- Understand the ALMA proposal review criteria (see Slide 7)

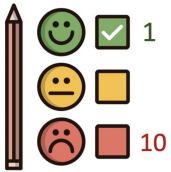
➤ Review Phase

- Read each proposal thoroughly
- Verify that proposals comply with anonymization requirements
- Write constructive, clear, and polite reviews ("the proposal did not...")
  Ethics: AI tools may be used to refine reviews (e.g., grammar),
  but do **not** submit proposals to AI tools (privacy violation!)
- Rank proposals based on their scientific merit & suitability
- Ensure rankings accurately reflect the comments

Note: The score is computed from the 9 best ranks out of 10 (see Proposer's Guide)

Ethics

- Do not share proposals!
- Do not copy and submit



1 → Strongest
10 → Weakest

Edit tips
- Bullet points
- A blank line btw sections

Review structure and professional wording

1. **One or two short sentences summarizing the proposal science**

   Example: Spatially resolved studies of giant molecular clouds at cosmic noon are crucial to test star-formation laws and galaxy evolution models. This proposal aims to [...], which would provide [...].

2. **List of strengths (there are always strengths)**

   This is an interesting project, for which notably:
   - Strength #1
   - Strength #2

3. **List of weaknesses: be constructive, clear, and polite**

   That said, a few questions came to mind, not as criticisms, but as ideas to make the proposal even stronger. The proposal would have benefited from:
   - How to improve Weakness #1
   - How to improve Weakness #2

"Many similar targets are already known, such that it remains unclear why this is an interesting project."

"This is a modest time request, which has the potential to produce interesting results."

Proposal Results

Timeline

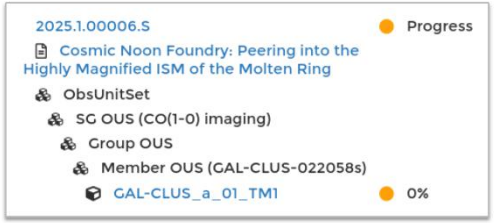
A	B	C	U
Highest priority	High priority	Backup projects	Rejected projects

- **Proposal priority grading**
  - (A) Highest priority programs; to be observed this Cycle or the next one
  - (B) May be observed this Cycle
  - (C) If A or B cannot be observed
  - (U) Rejected; improve and resubmit!
- **Selection considerations**
  - Score based on the 9 best reviews
  - Requested observing setup (e.g., bands, array configurations)
  - Science categories: similar OSR

Proposal Results

Timeline

- **SnooPI: Project Monitor**
  https://asa.alma.cl/snoopi
  - Visualize proposal feedback
  - Observing program status (after Phase II submission)

A	B	C	U
Highest priority	High priority	Backup projects	Rejected projects

SnooPI ➤ interface